Networking Basics: DNS (Domain Name System)

DNS (Domain Name System) is the internet's "phonebook." It translates human-readable domain names (like www.google.com) into IP addresses (like 142.250.190.78) that computers use to communicate.

Why DNS is Needed

People remember names more easily than numbers.

- Without DNS: You would need to type 142.250.190.78.
- With DNS: You simply type www.google.com.

DNS automatically finds the correct IP address.

How DNS Works

When you enter a website address in your browser:

1. You type www.example.com.
2. Your browser checks if the IP address is already cached.
3. If not, it asks a **DNS Resolver**.
4. The resolver contacts DNS servers to find the correct IP address.
5. The IP address is returned.
6. Your browser connects to the web server using that IP address.
7. The website loads.

Flow:

User

|

▼

Browser

|

▼

DNS Resolver

|

▼

DNS Servers

|

▼

IP Address Returned

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▼

Website Opens

Types of DNS Servers

1. DNS Resolver

- Receives DNS requests from your device.
- Usually provided by your ISP or a public DNS service.

2. Root DNS Server

- The first stop in the DNS lookup process.
- Directs the resolver to the correct Top-Level Domain (TLD) server.

3. TLD (Top-Level Domain) Server

- Manages domain extensions such as:
    - .com
    - .org
    - .net
    - .edu
- Points to the authoritative DNS server.

4. Authoritative DNS Server

- Stores the actual DNS records for a domain.
- Returns the correct IP address.

Common DNS Record Types

Record	Purpose	Example
A	Maps a domain to an IPv4 address	example.com → 192.168.1.1
AAAA	Maps a domain to an IPv6 address	example.com → 2001:db8::1
CNAME	Creates an alias for another domain	www.example.com → example.com
MX	Specifies mail servers	Gmail mail servers
NS	Identifies authoritative name servers	ns1.example.com
TXT	Stores text information (e.g., SPF, verification)	Domain verification

Example DNS Lookup

Suppose you visit www.google.com:

- Browser requests the IP address.
- DNS Resolver checks its cache.
- If not found:
    - Root Server → directs to .com TLD server.
    - TLD Server → directs to Google's authoritative DNS server.
    - Authoritative Server → returns the IP address.
- Browser connects to Google's server and loads the page.

Benefits of DNS

- Easy-to-remember website names.
- Fast access through DNS caching.
- Scalable for millions of websites.
- Reliable and distributed worldwide.
- Allows websites to change IP addresses without changing their domain names.

Public DNS Providers

- **Google Public DNS:** 8.8.8.8 and 8.8.4.4
- **Cloudflare DNS:** 1.1.1.1 and 1.0.0.1

- **OpenDNS:** 208.67.222.222 and 208.67.220.220

Interview Questions

1. What is DNS?
DNS (Domain Name System) translates domain names into IP addresses so computers can locate websites.

2. Why is DNS important?
It lets users access websites using easy-to-remember names instead of numeric IP addresses.

3. What is a DNS Resolver?
A server that receives DNS queries from clients and performs the lookup process.

4. What is an A Record?
A DNS record that maps a domain name to an IPv4 address.

5. What is the difference between A and AAAA records?

- **A Record:** IPv4 address
- **AAAA Record:** IPv6 address

6. What is DNS caching?
DNS caching stores previously resolved IP addresses temporarily, reducing lookup time and improving performance.

Key Points to Remember

- DNS = **Domain Name System**
- Converts **domain names → IP addresses**
- Uses a hierarchical structure:
    - Resolver
    - Root Server
    - TLD Server
    - Authoritative Server
- Common records: **A, AAAA, CNAME, MX, NS, TXT**
- DNS caching improves speed and reduces network traffic.